Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Two second-order streams create…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Second order

B. Fourth order

C. First order

D. Third order

Correct: D

2. (2 points) Hydrograph area represents…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Only peak stage

B. Only velocity

C. Only basin area

D. Volume

Correct: D

3. (2 points) Impervious connectivity can affect…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only mineral hardness

B. Only gene frequency

C. No water process

D. Runoff timing

Correct: D

4. (2 points) Groundwater divides always match surface divides?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Always

B. Only on maps

C. Only in winter

D. No

Correct: D

5. (2 points) A 100-year label means exact spacing?

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Always

B. Only for rivers

C. Only after one flood

D. No

Correct: D

6. (2 points) Load rate needs…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only pH

B. Only map area

C. Only color

D. Concentration and discharge

Correct: D

7. (2 points) Radial drainage can suggest…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Only flat sediment

B. Only a straight fault

C. No topography

D. An elevated central feature

Correct: D

8. (2 points) Stage and discharge are…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Identical units

B. Always equal numbers

C. Unrelated by any model

D. Related but different quantities

Correct: D

9. (2 points) Before/after change proves causality alone?

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Always

B. Only after one storm

C. Only in a reservoir

D. No

Correct: D

10. (4 points) Why specify the outlet?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

answer: It defines the contributing basin and network considered. (4 points).

11. (2 points) A freshwater catchment receives 130 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

answer: 60 mm (2 points).

Numerical target 60; absolute tolerance 0.12. Accepted units: , mm.

12. (2 points) A rectangular stream is 7 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

answer: 7 m^3/s (2 points).

Numerical target 7; absolute tolerance 0.014. Accepted units: , m^3/s.

13. (2 points) A river channel is 15 km long between points 11 km apart along the valley. Find sinuosity.

answer: 1.3636363636363635 (2 points).

Numerical target 1.3636364; absolute tolerance 0.0027272727. Accepted units: , .

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

answer: 0.02 (2 points).

Numerical target 0.02; absolute tolerance 4e-05. Accepted units: , .

15. (2 points) Use Q=K A i with K=0.001 m/s, A=23 m^2 and i=0.02. Find saturated Darcy discharge.

answer: 0.00046 m^3/s (2 points).

Numerical target 0.00046; absolute tolerance 9.2e-07. Accepted units: , m^3/s.

16. (2 points) A lake stores 3000 m^3 and exports 300 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

answer: 10 days (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , days.

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

answer: Faster runoff routing can shorten lag time and increase peak discharge; drainage infrastructure, storage and storm properties must be controlled. (2 points).

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

answer: Erosion is commonly concentrated on the outer cut bank and deposition on the inner point bar, with local flow conditions controlling details. (2 points).

19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

answer: Second order; joining two channels of the same order increases order by one. (2 points).

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

answer: Lower groundwater head can reduce discharge to the river or induce infiltration from it, depending on hydraulic connection and gradients. (2 points).